Senior High School

Lesson Exemplar in General Science

Quarter

LE

6

Lesson Exemplar for General Science
Quarter 2: Unit 3

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Republic of the Philippines

Department of Education

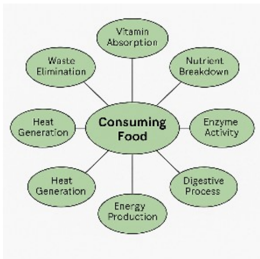
BUREAU OF LEARNING DELIVERY

LESSON EXEMPLAR			
Learning Area	General Science	**Grade Level**	11
Semester	1st	**Quarter**	2

I. OBJECTIVES *(Identifying the Goals)*	
Content Standard	The learners learn that chemical reactions are part of human activities and present in the environment.
Performance Standard	By the end of the quarter, learners apply research skills to find out how scientists have made an impact on our lives. They recognize the significant role of different agencies in the Philippines, such as DOST and DTI, in scientific innovation and development. They show critical thinking in explaining the different substances present in household and personal care products including their benefits and potential risk of using and disposing of them. They demonstrate appropriate and safe handling and use of chemicals. They explain that chemical reactions take place in human activities and in the environment, and that there are several chemical solutions that pose a threat to the community.
Learning Competencies	**Key Topics Learned in the Previous Grade Level:** 1. Properties and classification of matter (Grade 8 - Science) 2. Evidence of chemical reactions and balancing simple equations (Grade 8 - Science) 3. Cell structure and function, including the mitochondria (Grade 9 - Science) 4. Functions of carbohydrates, proteins, and fats in the body (Grade 9 - Science) 5. Interaction of digestive, circulatory, and respiratory systems (Grade 10 - Science) In this lesson, grade 11 learners describe the chemical reactions that take place in our body cells, which are referred to as metabolism, and explain their significance.
II. REFERENCES and MATERIALS	Gumaru, Michael. n.d. “Nutritional Guidelines for Filipinos: a prescription to good nutrition.” FNRI Website. https://www.fnri.dost.gov.ph/index.php/publications/writers-pool-corner/57-food-and-nutrition/204-nutritional-guidelines-

(*Selecting Resources and Material*)	for-filipinos-a-prescription-to-good-nutrition. University of Colorado Boulder. 2025. PhET Interactive Simulation: Cellular Respiration. https://phet.colorado.edu/en/simulation/legacy/cellular-respiration Health Chronicle. 2016. *How the Digestive System Works.* YouTube video, 3:36. June 3, 2016. https://www.youtube.com/watch?v=_T_vmcLyTzI OpenStax. 2016. Biology 2e. Houston, TX: OpenStax, Rice University. https://openstax.org/details/books/biology-2e Khan Academy. 2023. “Introduction to Metabolism.” Khan Academy. https://www.khanacademy.org/science/biology/energy-and-enzymes/metabolism/a/introduction-to-metabolism National Institute of General Medical Sciences (NIGMS). 2019. “Inside the Cell: Metabolism.” Bethesda, MD: U.S. Department of Health and Human Services. https://www.nigms.nih.gov/education/fact-sheets/Pages/inside-the-cell.aspx
(These shall be accomplished per topic)	
III. CONTENT *(Sequencing Content)*	**Metabolism as Chemical Reaction in the Body** • Reactants and Products in Metabolism • Significance of Metabolism • Health Application and Advocacy
IV. OBJECTIVES *(Setting Clear Objectives and Analyzing the Tasks)*	At the end of the lesson, the learners will be able to: 1. Define metabolism as the set of chemical processes that occur within living cells to maintain life. 2. Differentiate between anabolic and catabolic reactions and provide examples of each as they occur in the human body. 3. Identify how nutrients such as carbohydrates, proteins, and fats serve as reactants in metabolic chemical reactions. 4. Describe the products of metabolic reactions, such as energy (ATP), water, and other compounds needed by the body. 5. Explain the significance of metabolism in maintaining energy balance and supporting essential life

<table>
<tr><td></td><td colspan="2">processes such as growth, repair, and waste elimination, and relate this to safe practices in nutrition and chemical use.
6. Determine how agencies such as DOH, DOST, and local health initiatives support metabolic health, and use the findings to assess the role of metabolism in the body.
7. Promote safe practices in food and chemical use through a visual output such as a poster, graphic organizer, or video presentation.</td></tr>
<tr><td colspan="2">IV. PROCEDURES
(Selecting Strategies, Making Meaningful Content, Delivering Lesson and Assessing Learning)
This section focuses on selecting learner-centered, evidence-based instructional approaches such as problem-based learning, collaborative tasks, interdisciplinary integration, and technology-enhanced instruction. These strategies are intended to foster active engagement, critical thinking, and adaptability across diverse learning pathways. The chosen approaches and methodologies will be reflected through varied and relevant activities and assessments that emphasize real-world relevance and application, thereby enhancing learner engagement and comprehension.
(Each part shall have 2-3 varied activities)</td><td>ANNOTATION
*Instruction to teacher on how to facilitate the activities.
*In the Annotation, explicitly explain how the IDF is applied in each part of the lesson</td></tr>
<tr><td>A. Activating Prior Knowledge</td><td>1. Activating Prior Knowledge
Option 1: Think-Pair-Share
Objective: Recall daily tasks that require energy and connect them to internal body processes.
What You Need: Paper or Notebook and Pen, timer (optional)
What to DO:
1. Ask the learners to silently answer these questions for 3-4 minutes:
• What are your daily activities that require energy or</td><td>(Start by activating the learners' interest with familiar, concrete examples like eating breakfast or walking to school. Guide them if they struggle with questions by rephrasing or using relatable prompts: "When do you feel hungry or tired?" Records key words or phrases on the board (e.g., "energy," "food," "movement," "body process, etc.)
This activity sets the cognitive stage for new</td></tr>
</table>

	physical effort? • Why do you think energy is important in doing those daily activities? • What do you think happens inside your body that gives you that energy to do those daily activities? 2. Let them pair with a seatmate and share their thoughts about why energy is needed for those daily activities, and their idea of how the body produces that energy. 3. Each pair presents a summary of their discussion to the class. 4. Record key words or phrases on the board (e.g., "energy," "food," "movement," "body process"). **Option 2:** Metabolism Mind Map **Objective:** Express initial ideas about what happens to food inside the body. **What You Need:** Manila paper or chart paper, Colored markers or pens, Sample mind map (optional, for guidance) **What To Do:** 1. Divide the class into small groups with 3–5 members. 2. Ask each group to brainstorm and draw what they think happens inside the body after eating. 3. Encourage them to include food, parts of the body involved, possible changes, and outputs (e.g., energy). Encourage them to be creative. 4. Each group presents their mind map in 2-3 minutes. Sample map:  	*learning by drawing from students' daily experiences. It aligns with the SHS IDF's emphasis on relevance and responsiveness, bridging learners' prior understanding with the scientific concept of metabolism.* *(Before beginning, remind students there are no "wrong" answers—this is about what they already know or assume.* *If learners struggle to get started, guide them with prompts like: "What parts of the body are involved when you eat?" or "What do you think food becomes after digestion?"* *During group sharing, take note of misconceptions or oversimplifications. These will be useful for redirecting and refining understanding in the next phases. Note that if the internet is available and most of the students have laptop or smartphones, you can use mentimeter's word cloud instead.)* *This activity supports the SHS IDF through responsive instruction and diagnostic assessment, setting up meaningful engagement with new concepts.*

	(This image is created by the author) **Option 3**: Video Clip Screening **Objective:** Observe internal body changes after food consumption. **What You Need:** video clip, TV/projector or screen with audio, observation worksheet or blank paper, pen or pencil **What To Do**: 1. Have the students watch the 3D animation video titled "How the Digestive System Works" by Health Chronicle (https://www.youtube.com/watch?v=_T_vmcLyTzI) 2. While watching, they need to jot down observations about: • The sequence of organs the food passes through • The changes food undergoes in each organ • Any energy-related processes mentioned 3. Answer these Guide Questions as you do your note taking. • What changes did you observe inside the body after food consumption? • Which organs or systems were involved in these changes? • What do you think was the purpose of these changes? 4. Facilitate a short class sharing of observations, grouping	*(Before playing the video, prompt students to pay attention on how the food changes as it moves through each part of the digestive system. After viewing, facilitate a discussion by asking: "Why do you think these changes are necessary for our body?"* *If students focus only on mechanical aspects (like chewing), guide them to consider chemical processes by asking: "What happens to the food at the molecular level?")* *This activity aligns with the SHS IDF's emphasis on connecting prior knowledge with new information, fostering a deeper understanding of metabolism.* *(The teacher may provide a simple objective checklist to each student, allowing them to track their progress throughout the lesson. This promotes learner ownership, supports*

	similar answers together on the board. 2. Establishing the Purpose of the Lesson *Introduce first the learning objectives in verbatim manner, “At the end of the lesson, I will be able to (state the objectives).” **Option 1:** Food Beyond Energy: Freedom Wall **Objective:** Explain the significance of metabolism in maintaining energy balance and supporting life processes. **What You Need:** Sticky notes or index cards, manila paper, whiteboard, or designated “question wall” space Alternative: Digital tools that allow learners to post and organize their responses in real-time, such as Padlet, Jamboard, or Google Slides, may be used to simulate the “question wall” in an online or tech-integrated setup. **What To Do**: 1. Ask this guiding question: “Aside from giving us energy, what else does food do for our body?” 2. Give each student a sticky note or access to the digital board. 3. Instruct them to write 1–2 sentences responding to the question based on their prior knowledge or opinion.	*metacognition, and helps both teacher and student identify which objectives have been met and which still need reinforcement. This practice aligns with Reflective Instruction under the SHS IDF and encourages active goal setting.* *(This activity invites learners to think critically about the deeper functions of food beyond just "making us full" or giving energy.) If learners respond narrowly (e.g., "for strength"), ask follow-up questions like: “What happens when we’re sick or injured—does food help?” or “Do you think energy is the only thing our body needs from food?”* *Use their responses to transition smoothly into the next part of the lesson. Highlight how the class’s collective insights already touch on the core ideas of metabolism. This aligns with SHS IDF’s focus on learner-centered and reflective instruction, grounding the lesson in real-life relevance and internal motivation.)*

	4. Let them post their answers on the "Freedom Wall." 5. After all responses are posted, read selected entries aloud and categorize them (e.g., growth, repair, energy, waste removal and survival). 6. Ask the class: "What common theme or idea can you see from all these responses?" Let students suggest and agree on a shared theme. 7. Wrap up by summarizing the answers and using them to introduce metabolism as the body's way of converting food into energy and materials needed for life. **Activity 2**: Purpose Card **Objective**: Explain the significance of metabolism in maintaining energy balance and supporting essential life processes. **What You Need**: Small index cards or one-fourth sheets of paper, pen or pencil, Optional: collection box, folder **What To Do**: 1. Distribute a "Purpose Card" to each student. 2. Post this guiding question on the board: 3. Aside from giving us energy, what else do you think food does for the body?" 4. Instruct learners to reflect and write a 1–2 sentence answer. Encourage them to include why they think today's lesson is important. 5. Collect the purpose cards or let students post them in a designated area of the room. 6. Choose a few responses to read aloud and affirm, emphasizing the importance of food in supporting growth, healing, body functions, and overall health.	

	7. End by stating: “In this lesson, we’ll explore what really happens inside the body when we eat food—and how this helps keep us alive, healthy, and strong. That process is called metabolism.” **Activity 3**: Why Metabolism Matters: a Guided Class Discussion **Objective**: Explain the significance of metabolism in supporting life processes and link food functions to real-life concepts of health and wellness. **What You Need**: Prepared question prompts on the board or slide, whiteboard or manila paper, marker or chalk **What To Do**: 1. Post this guiding question: “How does learning about what happens to food inside the body help us stay healthy?” 2. Give the learners 1–2 minutes to reflect silently or jot down thoughts on the question. 3. Facilitate a class discussion by calling on volunteers to share their answers. 4. Use follow-up questions like: • “What happens if we eat too much or too little?” • “How could this knowledge help prevent sickness or improve performance in school?” 5. Record key responses on the board and group them into themes (e.g., energy, growth, recovery, disease prevention). 6. Ask the class: “What common theme or idea can you see from all these responses?” Let students suggest and agree on a shared theme. 7. Conclude the discussion with a synthesis:	*(This guided discussion activates learners’ curiosity and helps them connect the lesson to real-life contexts. Encourage learners to personalize their responses. For those who hesitate, ask questions like: “Have you ever skipped a meal? How did you feel afterward?”* *The open-ended format gives learners space to reflect, while your facilitation ensures direction. This aligns with the IDF principle of Reflective Instruction, helping learners see the relevance of metabolism to health, growth, and everyday choices.)*

	"All of your ideas point to one amazing body process called metabolism. Let's explore it together today and the next days."	
B. Instituting New Knowledge	1. Presenting Examples **Activity** 1: Catabolic vs Anabolic: Venn Diagram Objective: Differentiate anabolic and catabolic reactions and provide examples of each as they occur in the human body. **What You Need**: Printed or drawn Venn diagram templates (or blank paper if learners will create their own), Pens or colored markers, body processes written on strips or displayed on the board (digestion, cellular respiration, muscle growth, muscle breakdown protein synthesis, bone growth), reference materials or textbook (optional) **What To Do**: 1. Distribute a Venn diagram worksheet to each student or pair. Label one circle "Anabolic Reactions" and the other "Catabolic Reactions." 2. Review and explain each of the sample body processes (e.g., digestion, bone growth) briefly to ensure all learners understand them. 3. Ask learners to categorize each body process by placing it in the appropriate section of the diagram: *Left Circle* (Anabolic): for processes that build or synthesize complex molecules (e.g., muscle growth). *Right Circle* (Catabolic): for processes that break down molecules (e.g., digestion). *Overlapping Section*: for shared features of both anabolic and catabolic reactions (e.g., both are part of metabolism,	*(Begin with relatable examples (e.g., "When you grow taller, is that building or breaking?") to anchor new terms in familiar contexts.* *If learners struggle to categorize a process, guide them with prompts like: "Does this involve building something in the body?" or "Is energy being released or used here?"* *Use this task to correct misconceptions early while encouraging higher-order thinking through comparison. The visual and collaborative format aligns with the SHS IDF's responsive instruction and supports conceptual development within the Design phase of ADDIE.)*

	require enzymes, occur in cells, and contribute to energy balance). 4. Instruct learners to label their diagram clearly and include at least two examples for each category. Encourage the use of colored markers to distinguish categories. 5. Guide learners to discuss these key reflection questions in pairs: • What do anabolic and catabolic reactions have in common? • How are they different in terms of energy use or release? • Why are both necessary for the body to function properly? 6. Invite pairs or volunteers to share their diagrams and insights with the class. 7. Facilitate a brief class discussion, summarizing the roles of both types of reactions in metabolism, and address any misunderstandings using real-life examples. **Activity 2:** What's Happening in Their Body: A Case Study **Objectives:** 1. Differentiate anabolic and catabolic reactions and provide examples of each as they occur in the human body. 2. Identify how nutrients such as carbohydrates, proteins, and fats serve as reactants in metabolic chemical reactions. 3. Explain how lifestyle and diet influence metabolic health and energy balance. **What You Need:** printed or projected case scenario, guide questions handout or display on the board, writing materials or	*(This activity bridges real-life experiences and abstract scientific concepts by presenting a comparative scenario involving two individuals with contrasting lifestyles. It activates critical thinking and supports learners in identifying how nutrient intake and physical activity influence metabolic*

	group worksheet, and an optional graphic organizer to sort metabolic reactions Sample Case Scenario: Liam and Jordan are both 17-year-old Grade 11 students in the same class. Liam follows a balanced lifestyle. He eats at regular home-cooked meals with fruits, vegetables, lean proteins, and whole grains. He plays volleyball three times a week, sleeps 7–8 hours a night, and drinks plenty of water. Recently, Liam has noticed improved stamina and muscle tone. In contrast, Jordan skips breakfast, relies heavily on energy drinks and processed snacks, and often eats fast food late at night. He spends long hours on his phone or computer and rarely exercises. Lately, he has been feeling tired, weak, and has experienced unintentional weight loss. Medical tests reveal that his ATP levels are low, and his body is starting to break down muscle tissue to compensate for energy deficiency. **What To Do:** 1. Form groups of 3–5 learners. Assign each group to analyze both Liam and Jordan's case. 2. Read the case scenarios aloud or allow students time to read them silently. 3. Distribute or display the following guide questions and ask learners to answer and discuss: *Identifying Metabolic Reactions:* • What body processes are evident in Liam and Jordan's	*processes like anabolism and catabolism.* *Encourage group analysis, and if misconceptions arise—such as learners identifying muscle breakdown as anabolic—redirect by prompting:* *"Is the body constructing something new, or breaking something down for energy?"* *This scenario also opens opportunities for personal reflection. Ask students to connect the case to their own habits and decisions about food, rest, and activity. This aligns with the SHS IDF principles of Relevant and Responsive Instruction, and strengthens higher-order cognitive skills such as analysis, prediction, and self-assessment.)*

	lifestyle and physical condition? • Which of these processes are examples of anabolic reactions? Which are catabolic? • What nutrients (carbohydrates, proteins, fats) are likely involved in these processes? • How do their bodies likely differ in terms of energy production (ATP)? *Analyzing Nutrition and Lifestyle Impact:* • How does Liam's lifestyle support metabolic health? • What might happen to Jordan if his current lifestyle and eating habits continue? *Reflection and Personal Insight:* • Whose lifestyle do you relate to more—Liam or Jordan? Why? • Based on what you've learned about metabolism, what changes, if any, would you apply to your own habits? 4. Each group will share 1–2 key insights with the class. Clarify concepts and provide feedback after each presentation. 5. Facilitate a short class discussion highlighting how metabolism is influenced by both diet and physical activity. Reinforce how anabolic and catabolic processes must be in balance for optimal health. **Activity 3**: The Human Body as a Factory **Objective:** 1. Identify how nutrients such as carbohydrates, proteins, and fats	*(This activity makes abstract biochemical concepts more accessible through analogy. You can model the idea using a simple sketch first (e.g., delivery truck = bloodstream; assembly line = digestion). If*

	serve as reactants in metabolic chemical reactions. 2. Describe the products of metabolic reactions such as energy (ATP), water, and other compounds needed by the body. **What You Need:** Illustration of a basic factory (optional: printed or projected) worksheet, pens or markers, metacards for raw materials and products **What To Do:** 1. Introduce the analogy: "Imagine your body as a factory. What goes in? What happens inside? What comes out?" 2. Divide the class into small groups. Ask them to create a graphic organizer to represent the factory on how nutrients (raw materials) are processed into useful products like ATP and water and by-products like waste. 3. Encourage them to label parts of the "factory": • Raw materials: carbs, proteins, fats. • Machinery/processes: digestion, cellular respiration, enzyme actions • Products: ATP, amino acids, water, urea • Waste removal: sweat, urine, carbon dioxide (CO_2) 4. After completing their diagrams, let each group explain their factory model briefly to the class. 5. Synthesize by connecting each "part" of the factory to its biological equivalent in metabolism. 2. Discussing New Concept **Activity 1:** Understanding Metabolism: Class Discussion (In this part of the lesson the teacher will explain metabolism, anabolic and catabolic reactions using an interactive	*learners struggle to map biological equivalents, prompt them with: "Where would the mitochondria fit in this factory?" or "What part of the body removes waste, like a disposal chute?"* *This analogy supports SHS IDF's principle of meaningful and relevant instruction. It helps learners see metabolism as a system, building their conceptual foundation before diving into more complex reactions.* *(Keep your lecture interactive and chunked into short segments (5–7 minutes per concept). Use visuals to reduce cognitive load and clarify abstract ideas. If students*

	presentation.)	*struggle to differentiate anabolic and catabolic, revisit the examples using body-based scenarios.* *Use cold calling or voluntary Q&A to keep engagement high. Reinforce learning with gestures, drawings, or analogies. This structured teacher input phase aligns with the SHS IDF's Design phase, allowing for scaffolded, responsive delivery of new knowledge.* *(Interactive simulations engage visual and kinesthetic learners, deepening conceptual understanding of abstract biological processes. If internet access is limited, prepare an offline video or GIF demonstration in advance.*

	Key Concepts to be Covered: ➢ Definition of Metabolism ➢ Two Types of Metabolic Reactions ➢ Examples of Anabolic and Catabolic Reactions in the Human Body ➢ Role of Nutrients in Metabolism ➢ Production and Role of ATP ➢ The Importance of Metabolism for Life Processes – energy balance, growth, repair, and elimination of waste. **Guiding Questions for Discussion:** • How would you describe metabolism in simple terms? • What makes anabolic reactions different from catabolic ones? • Can you name one process in your body that builds molecules? One that breaks them down? • Why do we need energy in the first place? • What nutrient do you think your body uses first when you need energy? • Where does the energy your body uses come from? • What might happen if your body could not perform metabolism properly? **Activity 2:** Breaking Down Nutrients to Make Energy Using PHET Simulation **Objective**: 1. Identify how nutrients such as carbohydrates, proteins, and fats serve as reactants in metabolic chemical reactions. 2. Describe the products of metabolic reactions such as energy (ATP), water, and other compounds needed by the body.	*Circulate to assist groups and clarify terms like "mitochondria" or "ATP." If learners seem confused, redirect with a big-picture question: "What did the body get from the food?"* *This task promotes inquiry-based learning and aligns with SHS IDF's responsive instruction and the Development phase of ADDIE, bridging theory and process through interactive engagement.* PhET INTERACTIVE SIMULATIONS *(Source: https://images.app.goo.gl/3HpRPHXEQtB5AJg28)* *(Inform learners that you will be doing a concept review game to reinforce what they have learned so far. Launch the quiz using Kahoot or another quiz app. If tech is unavailable, do it in quiz bee format with individual or team points. After each question, pause to explain the correct answer and discuss why other options are*

	What You Need: Laptop or tablet per group (if available), internet connection, worksheet **What To Do:** 1. Introduce the goal: "Today, we will explore how nutrients are broken down in the body to produce energy in the form of ATP." 2. Launch the simulation or video and walk students through one round as a demonstration. 3. Distribute a guided worksheet with prompts like: • What nutrient is broken down in this process? • Where in the cell does this occur? • What products are formed? 4. Allow groups or pairs to explore the simulation. 5. Facilitate a brief discussion afterwards where learners share what they have observed and how the body transforms food into usable energy. **Activity 3:** Concept Check Game: Kahoot/Quiziz or quiz bee style formative check to clarify key terms and relationships.	*incorrect. Recognize top scorers but emphasize learning over competition.* *This activity serves as a formative assessment, helping you and your learners assess understanding before proceeding to application tasks. This game approach enhances engagement, aligns with responsive teaching under SHS IDF, and helps consolidate learning before moving on to mastery tasks.)* *(This task strengthens learners' ability to associate specific nutrients with their biological roles and metabolic products. It promotes recognition and recall while also reinforcing functional understanding. If students struggle, allow collaborative matching in pairs.)*

3. Developing Mastery

Activity 1: Matching Type

Objective:

1. Identify how nutrients such as carbohydrates, proteins, and fats serve as reactants in metabolic chemical reactions.
2. Describe the products of metabolic reactions such as energy (ATP), water, and other compounds needed by the body.

Sample:

Nutrient Type	Function	Metabolic End Product
A. Carbohydrates	1. Muscle repair	a. Amino acids and nitrogen waste
B. Proteins	2. Immediate energy supply	b. Glucose → ATP + CO_2 + H_2O
C. Fats	3. Long-term energy storage & insulation	c. Fatty acids + ATP + ketones

Activity 2: Metabolic Reaction Sort: a Guided Worksheet

Directions: Complete the diagram below by filling in the missing terms. Label whether the reaction is ANABOLIC or CATABOLIC.

(This activity reinforces understanding of metabolic reactions through visualization. It supports responsive instruction and targets the application level of Bloom's Taxonomy.)

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(Image generated using ChatGPT (OpenAI), 2025.)
<table>
<tr>
<td>Diagram 1: Muscle Growth
Reactants: ____________
Enzyme: ____________
Product: ____________
Reaction Type: ____________</td>
<td>Diagram 2: Digestion of Starch
Reactants: ____________
Enzyme: ____________
Product: ____________
Reaction Type: ____________</td>
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<tr>
<td>C. Demonstrating Knowledge and Skills</td>
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1. Finding Practical Application
Option 1: Agency Explorer
Objective: Determine how agencies such as DOH, DOST, and local health initiatives support metabolic health and use the findings to assess the role of metabolism in the body.

What You Need: If possible, to give time to students to conduct interview to the different agencies or barangay health officials, if not Internet-connected devices or printed materials on DOH, DOST, LGU health programs, Art supplies or can use Canva, PowerPoint or adobe),video editing app
What To Do:
1. Group learners into 3–5 members and assign or let them choose one agency: DOH, DOST, or a barangay health
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(This activity develops research and presentation skills while deepening learners' understanding of real-world applications of metabolism. Encourage groups to assign roles (e.g., researcher, designer, speaker) to ensure equal participation.
You may assign this as a take-home task to allow time for interviews or deeper investigation. If done in class, guide students toward credible sources and websites. For classes with limited internet access, prepare printed news articles or brochures from DOH, DOST, or local health offices.
For groups needing support, suggest starting with a simple query such as “DOH nutrition
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	office. 2. Ask each group to research or conduct an interview about the agency's initiatives related to diet, metabolism, or public health. 3. Using the interview data, instruct them to create an infographic visual output such as *poster, graphic organizer or video presentation* that summarizes the agency's role in promoting metabolic health, explains the connection to processes like nutrition, energy, and growth, encourages youth-friendly awareness and safe practices 4. Let groups present their output briefly in front of the class or through a gallery walk. 5. Provide feedback and encourage students to reflect on how these agencies affect their daily health choices. **Option 2**: Safe Use Campaign: **Objective:** Demonstrate understanding of safe practices in food and chemical use by creating a visual advocacy output—such as a poster, skit, or slogan—that promotes awareness and responsible behavior at home, in school, or the community. **What You Need:** Art supplies or digital tools **What To Do**: 1. Divide the class into groups with 3-5 members and assign or let them choose a format: slogan, skit, or poster. 2. Give them this campaign challenge: "Create a health awareness material that teaches your peers about safe practices in food and chemical use—at home, in school, or in	*program Philippines."* *This task supports Relevant and Responsive Instruction under the SHS IDF and aligns with the Evaluation and Application phases of ADDIE. It empowers learners to connect scientific content with public health initiatives and recognize the value of science in their local community.)* *(This activity builds advocacy, creativity, and scientific communication. It allows students to express their learning in diverse formats while applying science concepts to real-life issues.* *You may assign this as a take-home performance task or provide scaffolds (e.g., slogan templates, sample poster layouts) for classes that need more guidance. For lower-performing groups, suggest starting with questions like: "What risky food or chemical habits have you seen around you?"* *This aligns with the SHS IDF's principle of Relevant and Reflective Instruction, supporting the Application phase of ADDIE. It*

the community."

3. Guide them to focus on one or more of the following:
   - Proper food handling and storage
   - Safe use and disposal of household chemicals
   - Reading and understanding food/chemical labels
4. Give time for group planning and creation (in class or as homework).
5. Let each group present their output in a short showcase or gallery walk.

Rubrics for Grading activity 1 & 2

Criteria	4 – Excellent	3 – Satisfactory	2 – Developing	1 – Needs Improvement
Content Accuracy	All information is accurate and clearly explains the agency's role in metabolic health.	Most information is accurate with minor errors; main idea is clear.	Some information is inaccurate or missing; connection to metabolism is unclear.	Information is mostly incorrect or lacks relevance to metabolism.
Clarity and Organization	Ideas are logically organized and easy to follow. Headings and sections are well used.	Information is generally organized; some parts may need better structure.	Organization is inconsistent; hard to follow at times.	Information is confusing or disorganized.
Creativity and Design	Visuals are highly engaging, creative, and effectively support the message.	Design is clear and visually appealing.	Visuals are basic or minimally support the message.	Lack of effort in design; visuals are distracting or unclear.
Team Collaboration	All members contributed and showed excellent teamwork.	Most members contributed; collaboration was evident.	Uneven participation; some members contributed more than others.	Little evidence of teamwork or collaboration.
Presentation and Explanation	Group confidently presented and clearly explained their output and its relevance.	Group explained output with some confidence and clarity.	Presentation lacked clarity or completeness.	Group did not present or explanation was unclear or missing.

Option 3: Food Energy and You

Objective: Explain the significance of metabolism in maintaining energy balance and supporting essential life processes like growth, repair, and waste elimination.

What You Need: Nutrient Tracker template, Sample food guides

gives learners a platform to take ownership of their learning and promote community wellness.)

(This activity connects metabolism with daily life and promotes health awareness. It encourages learners to observe how nutrient intake and energy output are related.

If students struggle with identifying

	or serving size visuals, Calculator (optional, for estimating energy use), Pen/paper or digital document **What To Do:** 1. Distribute or display a Nutrient Tracker Log where students can record: • What they ate for breakfast, lunch, and dinner • Estimated portion size • Type of nutrient (carbohydrate, protein, fat) • Main physical activities they did that day 2. Ask learners to monitor and record this data for one full day (homework) or during school hours (if limited to in-class). 3. The next day, instruct learners to analyze their entries by answering: • Which of your meals gave you the most energy or helped you feel full the longest? What foods contributed to this? • Based on your tracker, which nutrients (carbohydrates, proteins, fats) were most common in your meals, and how do they relate to your energy use for the day's activities? 4. Allow them to present their reflections as a written paragraph, a short oral sharing, or visual summary (optional). 2. Making Generalization **Activity 1:** 3-2-1 Summary Write 3 things you learned, 2 interesting facts, and 1 question you	*nutrients, provide a guide showing common foods and their dominant nutrient (e.g., rice = carbohydrate, fish = protein). If learners report irregular meals, use this as a starting point for discussing balanced diets.* • *The task supports Reflective Instruction in the SHS IDF and aligns with the Application and Evaluation stages of ADDIE, as it promotes self-assessment and metacognition by asking students how their eating and activity habits support or affect their body's ability to grow, repair, or stay energized throughout the day, thereby contextualizing the science of metabolism.* • *Sample Nutrient tracker log:*

still have. (In this activity, instruct the learners to reflect on everything they have learned about metabolism, nutrients, and body processes, and then fill in the 3-2-1 table below with their own insights.)

3 Things I Learned	2 Interesting Facts I Discovered	1 Question I Still Have
1. ______________________	1. ______________________	______________________
2. ______________________	2. ______________________	
3. ______________________		

Activity 2: Concept Web

(Students create a web connecting nutrients, metabolism, and body functions.)

Activity 3: Metabolism Tweet

(Learners write a 280-character summary of what metabolism is and why it matters.)

Guide Questions:

1. What is metabolism in simple terms?
2. What important role does metabolism play in the body?
3. How does metabolism affect your energy, growth, or health?
4. Why should people understand how metabolism works?

3. Evaluating Learning

Written Quiz:

NUTRIENT TRACKER LOG

Name: ______________________

Date: ______________ Day of Week: __________

MEAL	FOODS	APPROXIMATE PORTION	MAIN NUTRIENT
Breakfast			
Lunch			
Dinner			
Physical Activities			
Physical Activities	______ ______	______ ______	______ ______

	Directions: Identify the correct term that best completes each statement. Write your answer on the blank before each number. 1. ____________ is the term for all chemical reactions that occur within a living organism to maintain life. 2. The type of metabolic reaction that builds complex molecules and requires energy is called ____________. 3. ____________ reactions break down molecules to release energy. 4. The molecule that stores and delivers energy for cellular activities is ____________. 5. The nutrient primarily used by the body as a quick source of energy is ____________. 6. The organelle known as the “powerhouse of the cell” where ATP is produced is the ____________. 7. ____________ is the process of breaking down starch into glucose. 8. Proteins are broken down into ____________ during digestion. 9. The government agency responsible for promoting science-based nutrition and health practices in the Philippines is ____________. 10. A safe practice when handling household chemicals is to always ____________ before using them. Reflection Essay: In 5–7 sentences, reflect on how your current lifestyle—such as your diet, physical activity, or sleep habits—affects your metabolism. What changes (if any) would you like to make to improve your metabolic health? Support your answer with concepts learned in class.	*(The teacher may opt to use google form for immediate feedback)* *(Make sure to give immediate feedback to the students and provide criteria in checking their output.)* *Possible Answers:* *1. Metabolism* *2. Anabolism* *3. Catabolic* *4. ATP (Adenosine Triphosphate)* *5. Carbohydrates* *6. Mitochondrion* *7. Digestion (or specifically: Hydrolysis, if you want a more scientific term)* *8. Amino acids* *9. Department of Health (DOH)* *10. Read the label (or "read instructions" is also acceptable)*

	4. Additional Activities **Activity 1:** Video Review Task (Students watch an educational video on metabolism and summarize it using a visual aid.) **Suggested Videos:** 1. What Is Metabolism? https://www.statedclearly.com/videos/what-is-metabolism/ Duration: 7 minutes Description: This animated video explains metabolism in simple terms, making it ideal for beginners. It defines metabolism as all the chemical reactions that keep organisms alive, and introduces anabolic and catabolic reactions through everyday examples. Clear visuals and real-world analogies help students grasp the basic concept of how cells use food to build and power the body. 2. Metabolism & Nutrition, Part 1: Crash Course Anatomy & Physiology #36 https://www.youtube.com/watch?v=fR3NxCR9z2U Duration: 10 minutes Description: This fast-paced and engaging Crash Course episode explores how nutrients are broken down to release energy and build body structures. It covers key metabolic terms, the role of ATP, and how our bodies use carbohydrates, fats, and proteins. Ideal for students who enjoy energetic and structured explanations with real-life applications.	*Rubric for Reflection* *5 – Discussion is scientifically explained consistent to the concepts and has no misconception.* *4 – Discussion is scientifically explained consistent to the concepts, but with minimal misconception.* *3 – Discussion is explained consistent to the concepts, but with misconceptions.* *2 – Discussion is not consistent to the concept.* *0 – No discussion*

	3. Introduction to Metabolism: Anabolism and Catabolism – Khan Academy https://www.khanacademy.org/science/ap-biology/cellular-energetics/cellular-energy/v/introduction-to-metabolism-anabolism-and-catabolism Duration: 9 minutes Description: This video breaks down the two main types of metabolic reactions—anabolism and catabolism—while emphasizing the role of enzymes and energy flow. It's ideal for visual learners who want to understand how small molecules are built up or broken down inside the body, with a focus on ATP and the importance of these reactions to overall health and function. **Activity 2**: Peer Tutoring (Assign advanced students to support peers struggling with concepts.) **Activity 3:** Local Expert Interview (Interview a health worker or teacher about common nutritional practices in the community.)	*(Guide Questions if this will be used:* • *How does the video define metabolism, and why is it important for living organisms?* • *What is the difference between anabolic and catabolic reactions based on the examples provided?* • *Can you name a process in your body that builds something (anabolic) and one that breaks something down (catabolic)?)* *Guide Questions:* • *What roles do carbohydrates, proteins, and fats play in metabolism according to the video?* • *How does the body use ATP, and why is it considered the “energy currency” of the cell?* • *Which part of the video helped you understand metabolism more clearly,*

		and why? *Guide Questions:* • *What are the main differences between anabolism and catabolism as explained in the video?* • *How do enzymes and energy (ATP) influence metabolic reactions in the body?* • *Why is understanding metabolism important for making informed decisions about health and nutrition?*
V. ASSESSMENT *(Assessing Learnings)*	*• Performance Task: Learners create an advocacy campaign (poster, infographic, or short video) that highlights what metabolism is, how it supports the body, and how to promote it through healthy practices. This task will be evaluated using a rubric based on clarity, accuracy, creativity, and relevance.* *• Summative Test: Administer a 15-item multiple-choice test to the students to assess their understanding of the lesson. (**Please see annex 1 for the summative test and performance task.)***	
VI. REFLECTION *(Feedback and Continuous*	*This section presents the key highlights and challenges encountered by both teachers and learners during the teaching-learning process throughout the unit. It also includes the adjustments made by the teacher to improve instruction.*	

Improvement)	

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Annex 1. Suggested Assessment

Direction: Read each question carefully and choose your answer from the given options.

1. Which of the following best describes metabolism?
   a. The breaking of bones.
   b. The production of cells.
   c. Only the digestion of food.
   d. All chemical reactions in the body.
2. Your science teacher explains that metabolism includes two types of reactions that either build or break molecules. Which of the following best describes an anabolic process?
   a. Breaking down fats for energy.
   b. Converting glucose to ATP.

c. Building muscle proteins.
d. Releasing stored heat.

3. During exercise, your body uses energy by breaking down nutrients. What type of metabolic process is responsible for this?
    a. Anaerobic
    b. Anabolic
    c. Catabolic
    d. Synthesis
4. Miguel read that food gives us energy. He wonders which nutrient is most often used as fuel. Which nutrient primarily serves as the body’s main energy source?
    a. Carbohydrates
    b. Fats
    c. Proteins
    d. Vitamins

5. During a conversation about digestion, your younger sister asks, “What really happens to the food we eat once it’s inside our body?” Which of the following best explains what happens to food during the digestive process, from a metabolic perspective?
    a. The food molecules are broken down into simpler forms the body can use.
    b. The food is immediately turned into energy without change.
    c. The food is eliminated as waste without being processed.
    d. The nutrients are stored in the stomach for future use.
6. During a science activity, you learned that the body produces ATP through respiration. How does ATP function in the human body?
    a. Regulates body temperature.
    b. Provides usable energy.
    c. Digests proteins.
    d. Stores waste.

7. Which organelle is responsible for producing the most ATP during metabolism?
    a. Chloroplast
    b. Mitochondrion
    c. Nucleus
    d. Ribosome
8. Warren injured his leg while riding bicycle and was advised to eat protein-rich food. Which nutrient helps in tissue repair and growth?
    a. Lipids
    b. Glucose
    c. Proteins
    d. Water

9. Which of the following is a direct product of cellular respiration?
   a. ATP
   b. Nitrogen
   c. Oxygen
   d. Glucose
10. Your group is preparing an advocacy campaign to promote safe practices in food and chemical use at home. As part of your research, you identified important habits to include in your poster. Which of the following best supports your advocacy?
    a. Transferring leftover chemicals into empty water bottles to reuse containers.
    b. Reading product labels and safety instructions before using any item.
    c. Storing cleaning agents near kitchen condiments for convenience.
    d. Using strong-scented cleaners in closed spaces to remove odors.
11. In your health class, your teacher explains how lifestyle choices impact metabolism. What could be a metabolic consequence of poor diet and no exercise?
    a. Balanced anabolic and catabolic processes.
    b. Enhanced metabolism and energy.
    c. Fat storage and low energy.
    d. Increased muscle growth.
12. Two students are having a debate. One says food alone builds muscle, the other says exercise is needed too. Which statement is most scientifically accurate?
    a. Protein alone builds muscle.
    b. Exercise breaks down all proteins.
    c. Carbs are more important than protein.
    d. Food and exercise together support growth.
13. A public awareness campaign was launched to promote health and chemical safety. Which slogan best supports safe and healthy metabolic practices?
    a. Skip meals to burn fat.
    b. Take supplements every day.
    c. Chemicals make you stronger.
    d. Eat smart, move more, live safe.

14. Your barangay invited you to share tips on boosting metabolism naturally. Which advice can you give to best supports metabolic health?
    a. Skip breakfast.
    b. Drink energy drinks.

c. Avoid all carbohydrates.
d. Eat balanced meals and stay active.

15. Which of the following best explains the role of metabolism?

a. It causes disease.
b. It replaces exercise.
c. It supports energy and repair.
d. It controls breathing only.

Answer Key:

ITEM	ANSWER	RATIONALIZATION
1.	D	Metabolism includes all biochemical processes, such as breaking down nutrients and building body structures, not just digestion.
2.	C	Anabolism involves constructing complex molecules like proteins, requiring energy input.
3.	C	Catabolic reactions break down molecules (e.g., glucose, fat) to release energy needed during physical activity.

4.	A	Carbohydrates are the body’s preferred and quickest source of energy, especially glucose.
5.	A	Digestion involves breaking down complex food into nutrients the body can absorb and use for energy and repair.
6.	B	ATP (adenosine triphosphate) acts as the main energy currency in cells, fueling bodily functions and processes.
7.	B	Known as the powerhouse of the cell, mitochondria produce ATP through cellular respiration.
8.	C	Proteins provide the building blocks (amino acids) needed for tissue repair and muscle development.
9.	A	The main goal of cellular respiration is to convert glucose into ATP for cellular energy.
10.	B	Reading labels helps ensure safe usage and prevents harmful reactions or misuse.
11.	C	Without proper nutrition and activity, the body stores more fat and has lower energy due to slowed metabolism.
12.	D	Muscle growth requires both nutrients (especially protein) and physical stimulus like exercise.
13.	D	This encourages balanced nutrition, physical activity, and safety for better health and metabolism.
14.	D	Balanced nutrition and regular exercise naturally enhance metabolic efficiency.
15.	C	Metabolism provides energy for daily activities and supports the body's growth and maintenance.

Performance TASK: "*MetaboPower*: Advocating for a Healthy and Energized Life"

Task Description:

You are tasked to create an advocacy campaign material that aims to inform and inspire others to understand metabolism and promote healthy practices that support metabolic health. You may choose one format for your campaign material from the options below. You may use either English, Filipino or the local language in the material to allow your community to learn from your material.

- A poster (digital or hand-drawn)
- An infographic
- A 1–2-minute short video

Your campaign must clearly show:

1. What metabolism is (a clear and accurate definition)
2. How metabolism supports the body (e.g., energy, growth, repair, waste elimination)
3. At least two healthy practices that help maintain or improve metabolic health

Steps to Follow:

1. Research the topic using class materials, credible websites (e.g., DOST, DOH), or interviews.
2. Choose your medium and plan your design or script.
3. Create your advocacy material, ensuring it is informative, creative, and age-appropriate for your audience.
4. Submit your final output on the scheduled date.

Suggested Rubrics:

Criteria	**5 – Excellent**	**4 – Proficient**	**3 – Developing**	**2 – Beginning**
Clarity of Message	Message is very clear, engaging, and easily	Message is clear and understandable	Message is somewhat clear but may need more	Message is unclear or confusing

	understood		explanation	
Scientific Accuracy	Definition and examples are accurate and well-explained	Mostly accurate with minor errors	Some inaccuracies or missing scientific concepts	Mostly inaccurate or scientifically weak
Creativity and Presentation	Highly creative and visually appealing; strong effort is evident	Visually neat and creative; effort is evident	Minimal design or creativity; may look rushed	Lacks creativity; poorly presented
Relevance and Advocacy	Strong connection to real-life practices; encourages action	Shows good relevance and purpose	Partially relevant; advocacy not emphasized	Lacks relevance; no clear advocacy message